

Variable Voltage Capacitor — Solution

W23 (2023) — English translation

A1

0.30 pts

Initially, the charge on the capacitor is zero. Let at time $t = 0$ a constant voltage $V = V_0$ be applied to the input of the circuit. Find the dependence of the voltage on the capacitor on time $U(t)$ at $t > 0$.

Instantaneous value of the current through the capacitor

$$I = \frac{V_0 - U}{R}.$$

Ток equals производной chargea capacitора $I = \frac{dq}{dt}$, а charge можно выtimesандть via/through voltage. $q = CU$, hence $I = C \frac{dU}{dt}$, therefore equation на voltage

$$\frac{dU}{dt} = \frac{1}{RC} (V_0 - U),$$

and its solution

Answer: \textbf{Answer:}

$$U = V_0 (1 - e^{-t/RC}).$$

A2

0.20 pts

Find the dependence of the current through the resistor on time $I(t)$ ($t > 0$).

$$I = C \frac{dU}{dt}$$

Answer: \textbf{Answer:}

$$I = \frac{V_0}{R} e^{-t/RC}.$$

A3

0.20 pts

Find the heat Q_0 that will be released on the resistor during the entire process.

During the entire time, charge $q = CV_0$ passes through the circuit, on which work $A = V_0 q = CV_0^2$ is performed. The final energy of the capacitor is $W = CV_0^2/2$, and the heat is $Q = A - W$

Answer: \textbf{Answer:}

$$Q = CV_0^2 - \frac{CV_0^2}{2} = \frac{CV_0^2}{2}.$$

A4

0.60 pts

Let now the initial voltage on the capacitor at $t = 0$ be equal to U_1 , and as before, a constant voltage $V = V_0$ is applied to the input. Find the dependence of the voltage on the capacitor on time $U(t)$ and the current through the resistor $I(t)$.

The voltage equation is the same as in the first paragraph, and its solution, taking into account the initial conditions

Answer: \textbf{Answer:}

$$U = V_0 - (V_0 - U_1)e^{-t/RC}, \quad I = \frac{V_0 - U_1}{R}e^{-t/RC}.$$

A5

0.70 pts

Find the amount of heat Q_1 that will be released during the entire process from point \textbf{A4}. Construct (qualitatively) a graph of Q_1 versus U_1 .

During charging, a charge $q = C(V_0 - U_1)$ flows through the circuit, a change in the energy of the capacitor $\Delta W = \frac{CV_0^2}{2} - \frac{CU_1^2}{2}$, the work of the source $A = V_0q$, so the released heat is

$$Q = A - \Delta W = CV_0^2 - CV_0U_1 - \frac{CV_0^2}{2} + \frac{CU_1^2}{2} = \frac{C(V_0 - U_1)^2}{2}$$

Answer: \textbf{Answer:}

$$Q = \frac{C(V_0 - U_1)^2}{2}.$$

B1

0.30 pts

Let a long time pass. Find an expression for the current through resistor $I(t)$.

The charge of the capacitor after a long time is $q = CU \approx CV = C\alpha t$, and the current is $I = dq/dt$

Answer: \textbf{Answer:}

$$I = C\alpha.$$

B2

0.20 pts

Find the amount of heat Q that will be released on the resistor over a sufficiently long time t .

Power after process establishment $P = I^2R \approx C^2\alpha^2R$

Answer: \textbf{Answer:}

$$Q = C^2\alpha^2Rt.$$

B3

0.10 pts

Find the current through resistor I . Express it in terms of the instantaneous voltage across the capacitor U , the input voltage V and the circuit parameters.

Answer: \textbf{Answer:}

$$I = \frac{V - U}{R}.$$

B4

0.10 pts

Express the rate of change of voltage across capacitor dU/dt in terms of the current through resistor I and the circuit parameters.

Answer: \textbf{Answer:}

$$\frac{dU}{dt} = \frac{1}{C}I.$$

B5**0.40 pts**

Having differentiated the equation from \textbf{B3}, obtain a differential equation for the current through the resistor, that is, express dI/dt in terms of I , the circuit parameters and the constant α .

Answer: \textbf{Answer:}

$$\frac{dI}{dt} = \frac{1}{R} \left(\frac{dV}{dt} - \frac{dU}{dt} \right) = \frac{1}{R} \left(\alpha - \frac{I}{C} \right)$$

B6**0.10 pts**

What is the current through the resistor at the initial time $I(0)$?

Answer: \textbf{Answer:}

$$I = 0.$$

B7**0.50 pts**

Using the analogy of charging a capacitor, find the current through the resistor as a function of time $I(t)$.

From the equation it is clear that the current increases, as does the voltage across the capacitor when charging it, with the maximum current value $C\alpha$, and the characteristic time of current change $\tau = RC$, as when charging the capacitor.

Answer: \textbf{Answer:}

$$I = C\alpha \left(1 - e^{-t/RC} \right)$$

B8**0.70 pts**

Find the dependence of the voltage on the capacitor on time $U(t)$. Construct a (qualitative) graph of voltage versus time.

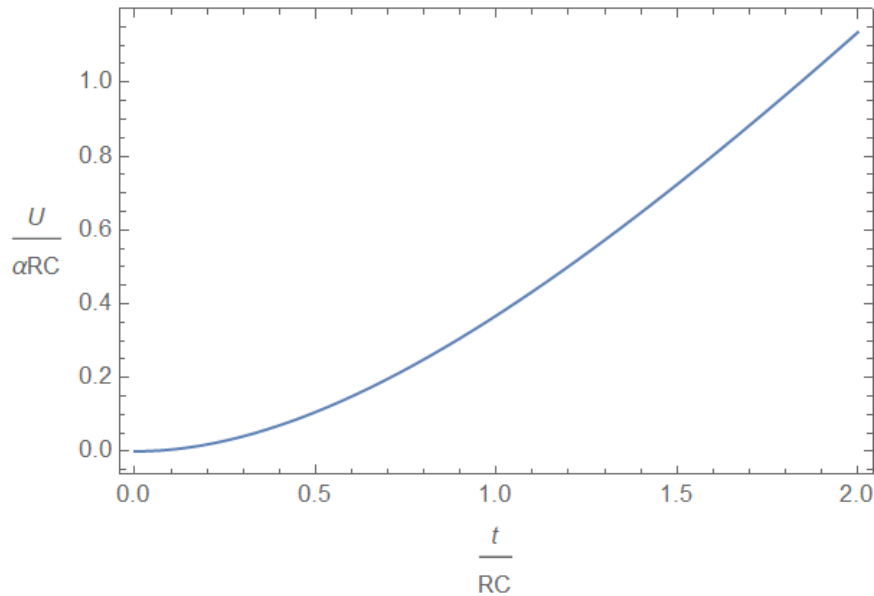
The charge on the capacitor can be found by integrating the current (with the initial charge being zero):

$$q = \int_0^t I(t)dt = C\alpha \left(t - RC \left(1 - e^{-t/RC} \right) \right) = C\alpha \left(t - RC + RCe^{-t/RC} \right),$$

a voltage $U = q/C$

Answer: \textbf{Answer:}

$$U = \alpha t - \alpha RC \left(1 - e^{-t/RC} \right).$$



B9

0.50 pts

Find the amount of heat $Q(t)$ that will be released on the resistor during time t .

The resistor releases power $P = I^2(t)R$, integrating it, we find the released heat

$$Q = \int_0^t I^2(t)R dt = C^2 \alpha^2 R \int_0^t \left(1 - e^{-t/RC}\right)^2 dt$$

Answer: \textbf{Answer:}

$$Q = C^2 \alpha^2 R \left(t - 2RC \left(1 - e^{-t/RC}\right) + \frac{RC}{2} \left(1 - e^{-2t/RC}\right) \right)$$

B10

0.30 pts

Find the voltage across the capacitor as a function of time at $t > T$.

This task is similar to the task from point A4. Here, as the initial voltage, you need to use the voltage at time T . You also need to use the $V_0 = \alpha T$ ratio. Then $V_0 - U_1 = \alpha RC (1 - e^{-T/RC})$.

Answer: \textbf{Answer:}

$$U = V_0 - \alpha RC \left(1 - e^{-T/RC}\right) e^{-\frac{t-T}{RC}}.$$

B11

0.30 pts

Find the amount of heat Q_2 that is released on the resistor at $t > T$.

This amount of heat can be calculated using the result from A5.

$$Q_2 = \frac{C(V_0 - U_1)^2}{2} = \frac{1}{2} \alpha^2 R^2 C^3 \left(1 - e^{-T/RC}\right)^2.$$

Answer: \textbf{Answer:}

$$Q_2 = \frac{1}{2} \alpha^2 R^2 C^3 \left(1 - e^{-T/RC}\right)^2.$$

B12**0.40 pts**

Find the amount of heat Q_T that was released on the resistor during the entire charging time of the capacitor.

We add up the answers from B9 and B11, present similar terms, and replace $\alpha = V_0/T$.

Answer: \textbf{Answer:}

$$Q_T = \frac{RC}{T} CV_0^2 - CV_0^2 \frac{R^2 C^2}{T^2} (1 - e^{-T/RC}).$$

B13**0.60 pts**

Find the ratio $k = Q_T/Q_0$. Here Q_0 is the heat that is generated at the resistor when voltage V_0 is instantly applied to the input of the circuit (see paragraph \textbf{A3}), Q_T is defined in the previous paragraph. Write down an approximate expression for k at $T \gg RC$.

$Q_0 = CV_0^2/2$, taking into account this

$$k = \frac{2RC}{T} - \frac{2R^2 C^2}{T^2} (1 - e^{-T/RC}).$$

На больших временах второе term много меньше первого,

$$k \approx \frac{2RC}{T}.$$

Note also, что for/at маленьких временах

$$e^{-T/RC} \approx 1 - \frac{T}{RC} + \frac{T^2}{2R^2 C^2} - \frac{T^3}{6R^3 C^3},$$

therefore

$$k \approx 1 - \frac{T}{3RC}.$$

Answer: \textbf{Answer:}

$$k = \frac{2RC}{T} - \frac{2R^2 C^2}{T^2} (1 - e^{-T/RC}) \approx \frac{2RC}{T}.$$

C1**0.20 pts**

Let a sufficiently long time pass and the vibrations become established. How should U_1 and U_2 be connected?

In steady state, opposite values of the external voltage correspond to opposite voltage values on the capacitor.

Answer: \textbf{Answer:}

$$U_2 = -U_1.$$

C2**0.20 pts**

What is the current through resistor I_1 when the input voltage is V_0 ?

Answer: \textbf{Answer:}

$$I_1 = \frac{V_0 - U_1}{R}.$$

C3

0.40 pts

By analogy with point B4, write an equation that is satisfied by the current through the resistor in the section where the input voltage decays linearly.

The input voltage in the section under consideration is

$$V = V_0 - \frac{4V_0}{T}t.$$

In equation for current enters only dV/dt , therefore constant term does not influence. Then the equation coincides with equation from part B5 with $\alpha = -4V_0/T$.

Answer: \textbf{Answer:}

$$\frac{dI}{dt} = \frac{1}{R} \left(-\frac{4V_0}{T} - \frac{I}{C} \right)$$

C4

0.50 pts

Taking into account the previous two points, find the dependence of the current on time in the section where the input voltage decays linearly. For ease of notation, consider that this section begins at time $t = 0$.

After a long time, the current will be equal to $I_2 = -4CV_0/T$, and the general solution of the equation has the form

$$I = I_2 + (I_1 - I_2)e^{-t/RC}$$

Answer: \textbf{Answer:}

$$I = -\frac{4CV_0}{T} + \left(\frac{V_0 - U_1}{R} + \frac{4CV_0}{T} \right) e^{-t/RC}$$

C5

0.60 pts

Using the found expression for the current, express the voltage on the capacitor at the end of the linear section U_2 in terms of the voltage at the beginning U_1 and the parameters of the problem.

Change in charge on the capacitor during voltage decrease

$$\Delta q = \int_0^{T/2} I(t)dt = -2CV_0 + RC \left(\frac{V_0 - U_1}{R} + \frac{4CV_0}{T} \right) (1 - e^{-T/2RC}),$$

then change in voltage

$$\Delta U = \frac{\Delta q}{C} = U_2 - U_1 = -V_0 - U_1 + \frac{4RC}{T}V_0 - \left(V_0 - U_1 + \frac{4RC}{T} \right) e^{-T/2RC}.$$

Answer: \textbf{Answer:}

$$U_2 = -V_0 + \frac{4RC}{T}V_0 - \left(V_0 - U_1 + \frac{4RC}{T} \right) e^{-T/2RC}.$$

C6

1.00 pts

Find the maximum voltage across capacitor U_1 at steady state. Write approximate expressions for U_1 for $T \ll RC$ and $T \gg RC$.

Solving the equation $U_2 = -U_1$ for U_1 we find (for convenience of writing, we use the so-called hyperbolic function $\tanh x$; if desired, the answer can be written simply in terms of exponentials.) For large values of the argument $\tanh x \approx 1$ and $U_1 \approx V_0 (1 - 4RC/T) \approx V_0$,

for small values of the argument $\tanh x \approx x - x^3/3$, $U_1 \approx \frac{T^2}{48R^2C^2}V_0$. If the answer to the behavior of a function cannot be obtained analytically, this can also be done numerically by calculating values for small values of time T/RC . So we can verify that at small times $U_1 \sim T^2$.

Note that the value U_1 is not actually the maximum voltage value across the capacitor, since the capacitor continues to charge some time after the input voltage has begun to decrease. The maximum voltage value is reached when the current through the capacitor becomes 0, after which the sign of the current changes and the capacitor begins to discharge. This time t_1 can be found from the condition

$$I(t_1) = 0 = -\frac{4CV_0}{T} + \left(I_1 + \frac{4CV_0}{T}\right)e^{-t_1/RC},$$

therefore

$$e^{-t_1/RC} = \frac{4CV_0/T}{I_1 + 4CV_0/T}, \quad I_1 = \frac{V_0 - U_1}{R} = \frac{4CV_0}{T} \tanh \frac{T}{4RC}.$$

Over time t_1 в capacitor затечет дополнительный charge

$$\Delta q_1 = \int_0^{t_1} I(t)dt = -4CV_0 \frac{t_1}{T} + RC \left(I_1 + \frac{4CV_0}{T}\right) (1 - e^{-t_1/RC}).$$

Substituting во второе сложение expression for экспоненты and напряжения U_1 we get

$$\Delta q_1 = -4CV_0 \frac{t_1}{T} + RC I_1.$$

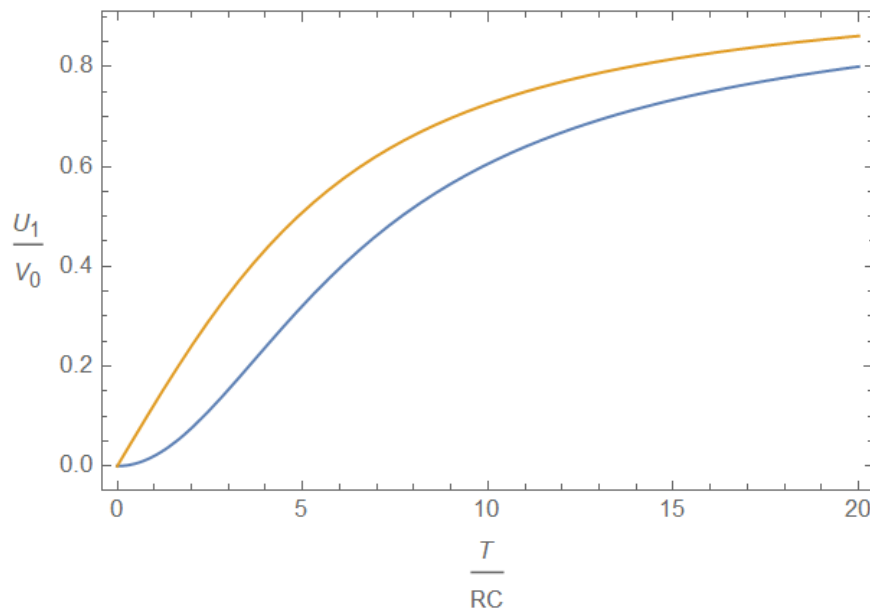
Then максимальное voltage

$$U_{\max} = U_1 + \frac{\Delta q_1}{C} = V_0 - 4V_0 \frac{t_1}{T} = V_0 \left(1 - \frac{4RC}{T} \ln \left(1 + \tanh \frac{T}{4RC}\right)\right).$$

Это voltage больше U_1 , for small in time values it depends linearly on time.

Answer: \textbf{Answer:}

$$U_1 = V_0 \left(1 - \frac{4RC}{T} \tanh \frac{T}{4RC}\right), \quad \tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$



C7

0.60 pts

Plot a graph of the ratio U_1/V_0 versus the dimensionless variable T/RC .

